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Training alters joint power distributions during running in National Football League Draft Preparation Players

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ABSTRACT
The aim of this study was to compare joint-specific contributions to total lower limb joint power during a speed run in American style football players before and after a 6-week National Football League (NFL) draft preparation camp. Fifty-two players performed one 5-second treadmill run at 5.5–6.5 m/s before and after camp. Speed was consistent between tests. Bilateral hip, knee, and ankle power generation (positive) and absorption (negative) were calculated from kinematics and kinetics using inverse dynamics. Negative total limb power decreased for both limbs, significantly for the left ($Z = -2.113$, $p = .035$, $r = .30$) due to significant decreases in negative ankle power bilaterally (left, $Z = -1.967$, $p = .049$, $r = .27$; right, $Z = -2.623$, $p = .009$, $r = .36$). The contribution of negative left hip power significantly increased ($Z = -2.398$, $p = .016$, $r = .31$). While no other significant changes in positive or negative joint-specific magnitudes or contributions occurred, there were general distal-to-proximal shifts in joint power contributions. Players appear to have adopted more energy efficient running through moderate joint-specific mechanistic training adaptations such as a reduction in ankle power absorption.

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KEYWORDS
American style football; team sport running; kinetics; performance; joint power

Introduction

Sprint speed is a key skill for American style football players. Players seeking to be drafted by the National Football League (NFL) showcase their linear speed running abilities in the 36.6 m (40 yard) dash at an annual combine. The combine is a weeklong invitation-only event where top university prospects (approximately 300) perform physical and mental assessments for personnel of the league's 32 teams. Research shows drafted players perform better in the 36.6 m sprint than non-drafted players (Sierer et al., 2008) and superior performance is associated with higher draft position (Tucker & Black, 2023). Combine invitees and other eligible players attend specialised training camps leading up to the NFL draft to prepare. Linear speed mechanics is one focus of these camps since players must transition from game play running to test performance running. Mokha et al. (2021) showed that 36.6 m time significantly improved in a group of American style football players participating in a six-week NFL draft preparation camp. Players appeared to accomplish this through decreasing ground contact time (pre, 177 ± 3.2 vs. post, 168 ± 2.2 ms) and increasing average vertical ground reaction force

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(vGRF) (pre, 1.5 ± 0.3 vs. post, 1.6 ± 0.1 BW). These changes represent hallmarks of higher velocity runs (Colyer et al., 2018; Morin et al., 2011; Weyand, 2017; Weyand et al., 2010). While these changes alone would not be enough to improve the participants' linear speed in the 36.6 m dash, they shed light on how foot-ground kinetics are modifiable in elite American style football players and provide foundation for further kinetic analysis.

During running, mechanical work must be done to rotate limb segments, restore energy lost during early ground contact, and propel the body forward. Given that ground contact times for sprinting range from 85 to 190 ms (Mattes et al., 2021; Numella et al., 2007; Weyand et al., 2010), the ability to perform work in a short time is vital and gives rise to examining joint power. Decrements in the capacity of the muscles to produce adequate power during running have been shown to negatively impact sprint speed (Sánchez-Sánchez et al., 2018; Small et al., 2009). Joint power is distinguished from system power where system power represents the rate at which the mechanical work is done on an external load and may veil the power contributions from the individual joints (Moir et al., 2012). Joint powers represent the rate at which mechanical work is accomplished at the joints providing estimates of muscle activity (Pires et al., 2014; Riddick & Kuo, 2022). Joint power magnitudes have been shown to increase as running speeds increase (Belli et al., 2002; Khassetarash et al., 2020). While improvements in speed and system power performance have been documented in American style football players undergoing training (Crawley et al., 2024; Gillen et al., 2024; Hoffman et al., 2004, 2009; Moore & Fry, 2007), the joint level mechanisms underlying adaptation have not been explored.

Positive joint power is primarily generated by concentric muscle contractions and potential strain energy is released. Most of the positive power (>60%) is generated at the ankle joint during ground contact when the plantar flexors (gastrocnemius and soleus muscles) contract to propel the body forward (Swinnen et al., 2021). Sanno et al. (2018) reported that prolonged running at near maximal effort shifted the distribution of positive joint work proximally in recreational but not competitive distance runners suggesting that shifting positive joint work from the ankle plantar flexors towards hip extensors could be detrimental to running economy. This shift has also been reported to occur in middle-distance runners completing an exhaustive run (181 ± 42 s to exhaustion) and in older runners as a sign of performance decline (Kim & Park, 2022). Furthermore, hip muscle activation during the swing phase consumes energy, leading to the estimated relative contribution of leg swing to the energetic cost of 7–11% (Swinnen et al., 2021). While these findings appear to indicate that a distal-to-proximal shift may be deleterious, it may be a necessary adaptation to faster running speeds. For example, the hip's contribution to average positive power in both stance and swing phases has been shown to significantly increase with faster running speeds (i.e., 6.0–9.0 m/s) (Schache et al., 2011, 2015). AminiAghdam et al. (2021) found that anterior trunk lean shifted power from the knee to the hip during level and uneven running at speeds of 3.5 m/s.

Negative joint work and power are primarily generated by eccentric muscle contractions where potential strain energy is stored and mechanical energy is reduced (DeVita et al., 2007). Since eccentric muscle contractions are associated with greater muscle damage (Hashizume et al., 2017), negative joint power is of particular interest. Negative power is

produced during the braking phase of running and is controlled reflexively by the leg extensors (Belli et al., 2002). The knee and ankle joints contribute greatly to limb power absorption compared to the hip joint (Hashizume et al., 2018). Belli et al. (2002) found the overall negative work absorbed at initial contact was lower than the positive power produced at end contact point, especially at the knee joint. The elongation time of the ankle and knee muscles is short and has been attributed to high ankle joint stiffness preventing yielding of the centre of mass at initial contact (Belli et al., 2002). It is not known how participating in a performance preparation program affects this mechanism.

Magnitudes and percentage contributions of joint power to total limb power may be sensitive to training aimed at improving linear speed mechanics even in highly trained athletes. Exploring joint level mechanics in American style football players may provide insight into the mechanistic adaptations of running performance that may carry over to other team sports where running is integral such as rugby, football (soccer), or Australian rules football. Therefore, the purpose of this study was to assess the changes in joint power magnitude and distribution during a high-speed run before and after a six-week training camp in American style football players. We hypothesised that speed focused exercises presented in training camp would redistribute the power contributions from the lower limb joints, characterised by a reduction of power from the ankle even if treadmill speed were kept uniform. To test the hypothesis, we used an inverse dynamic approach to calculate individual joint and total limb powers during a high-speed run before and after participation in training camp.

Materials and methods

Experimental approach

This study was part of a larger study monitoring pre-post changes in body composition, and vertical jump and speed running mechanics over the duration of a six-week training camp designed to prepare eligible American style football players for the NFL draft. Camp was divided into three microcycles that included drill specificity training, player position specificity training, resistance training and recovery-based movement (e.g., pool, soft tissue massage). The study procedures were conducted at the start of training camp and six weeks later at the end of training camp. We obtained written informed consent that was approved by the university's Institutional Review Board (#2018-684) and the study complied with the Declaration of Helsinki (World Medical Association, 2013).

Participants

A sample of convenience of 75 adult American style football players were recruited from a group of players enrolled in a specialised training camp at a local performance centre for the 2024 NFL draft. Players represented a variety of field positions (e.g., wide receivers, tight ends, defensive linemen) and had competed for different university teams in the United States. All participants had just completed their collegiate football season, were active players training 5-6× per week, and were cleared by licenced medical staff to participate in the study. All 75 completed the testing, but 23 were excluded due to

not achieving treadmill running speed of at least 5.5 m/s for trial duration ($n = 11$), retroreflective markers falling off during either the pre- or post-test ($n = 7$), or trials that qualitatively were deemed an unnatural running pattern by the investigators ($n = 5$). Data for 52 participants (age, 23.0 ± 0.9 yrs.; height, 1.85 ± 0.06 m; mass, 100.1 ± 19.6 kg) are presented in this study.

Protocol

Participants reported to the Sports Performance and Gait Laboratory in groups of three at a designated time over 2 ½ days at both the start and end of the six-week training camp. The training camp was divided into three microcycles that included drill specificity training, player position specificity training, resistance training and recovery-based movement (e.g., pool, soft tissue massage). Table 1 presents specific drills used by the coaches to improve linear speed mechanics. They were performed four times per week. The repetitions varied with relevance to other training demands of a session but were typically 5–10.

During data collection, participants wore compression shorts and their own running shoes. After explaining the study and procuring written informed consent, anthropometric measures were obtained according to the specifications of Vicon Nexus Plug-in Gait lower body model (Vicon, Centennial, CO, USA). These were used to estimate segment masses and moments of inertia necessary for inverse dynamics. To increase accuracy, the researchers used a felt tip marker to identify the anterior superior iliac spines since these landmarks are used for pelvis width, leg length, and marker placement. Following anthropometrics, participants completed a standardised 25-minute warm-up led by a single coach consisting of dynamic stretching, muscle readiness and reactivity exercises. Then, sixteen 14 mm retroreflective markers were placed bilaterally on the participant's pelvis and lower limbs according to the specifications of Vicon's Plug-in Gait lower body model and a local calibration was performed with the participant standing in anatomical position. The running trial took place on an instrumented split belt treadmill (Bertec Corporation, Columbus, OH, USA) with participants running on one side with zero incline. Participants began by walking on the treadmill at 1.0 m/s for

Table 1. Linear speed drills* used during the 6-week training camp.

Drill	Description
Ankling to butt kick series	Push toe to top of sneaker; lock ankle; hands break waste/hip; tuck head; move forward 4.6 m (5 yd.); progress to additional 4.6 m (5 yd) spent emphasising heel to buttocks
A-run series	9.1 m (10 yd) build-up jog then ankling to butt kicking series
Moving claw series	Push toe to top of sneaker; butt kick then clear knee; relax quadriceps and attack ground; gradually add rapid 1–2 arm motion
Prowler (sled) sprint	Feet together; push toe to top of sneaker; lock ankle; knee-up piston action; knees in front of body; no butt kick; then accelerate forward pushing sled; sled weight increases over sessions
Linear acceleration wall drill	45 degree lean against wall/fence; feet together; push toe to top of sneaker; lift one knee then drive straight down; switch legs slowly then rapidly increase steps
Assisted sprint series	Pulley system between athlete and coach where coach pulls athlete forward after the run begins
3-part start technique	Drill to teach set up for the 3-point start required by NFL
Kneeling arm drill	Half kneeling position; head down; long, pendulum arm swing progressing to arm swing at jog pace then run pace then sprint pace

*Complete explanations and demonstrations found at https://bbcom.me/36Kgq6P.

2–3 minutes. Speed was increased 1 m/s in one second increments where 3-second pauses occurred at jogging (~3.5 m/s) and fast jogging (~5.0 m/s) speeds to facilitate comfort. When the participant reached preferred maximum speed (maximum speed allowed by the treadmill was 6.5 m/s), they ran for 1- to 2-s before a 5-s recording was taken. This capture time was selected to mimic most 36.6 m (40 yd) run durations at the NFL combine. At the completion of the trial, participants transferred their weight to the non-moving belt, and the running treadmill belt was decelerated to a stop. Mean running speed was 6.4 m/s and the speed achieved at the pre-test was replicated at the post-test. Only one running trial was collected at both testing sessions. Participants were instructed to wear their same shoes to the post-test, but this was not assured by the researchers.

Instrumentation

Running kinematic and kinetic data were captured in a laboratory using a 10-camera 3D motion analysis system (Vicon, Centennial, CO, USA) sampling at 120 hz and recorded synchronously with an instrumented split belt treadmill (Bertec Corporation, Columbus, OH, USA) collecting at 1000 hz. The motion analysis system was calibrated on the day of each data collection session to an accuracy defined by a 3D residual below of .20 mm. Lower limb length, pelvis and lower limb breadth, and total body mass were measured using a common fibreglass tape measure, anthropometers (Lafayette Instruments, Lafayette, IN, USA), and an InBody 270 multifrequency bioelectrical impedance device (InBody USA, Cerritos, CA, USA), respectively. Height was self-reported.

Data processing

We compared the joint power absorbed (negative) and generated (positive) including the percent distribution from the hip, knee, and ankle to total limb power between weeks one and six of training camp. Power was derived from kinematic and kinetic data using inverse dynamics with rigid body assumptions. Marker coordinates were labelled and gap filled, and joint angles were calculated using Vicon Nexus software (ver. 2.15). Since participants ran on one side of the split-belt treadmill, gait cycles were visually inspected to identify left and right ground contact and foot-off events in the software based on vertical ground reaction forces. This allowed for synchronisation of the GRF with the motion capture data. Marker trajectories and ground reaction forces were filtered with a low-pass third-order Butterworth filter with a cut-off frequency of 10 hz and 50 hz, respectively.

Figure 1 depicts the free-body diagram of the lower limb. Joint moments (M_{joint}) were computed in Vicon Nexus using the Newton-Euler approach of inverse dynamics. See Equations 1 and 2 where Equation 2 is specific to the knee.

$$M_{joint} = I_{com} \cdot \alpha_{com} \tag{1}$$

$$M_{knee} = I_{com} \cdot \alpha_{com} - R_{ankle_y} \cdot L_{leg} \cos(\theta) + R_{ankle_x} \cdot L_{leg} \sin(\theta) + mg \cdot L_{prox} \cos(\theta) - M_{ankle} \tag{2}$$

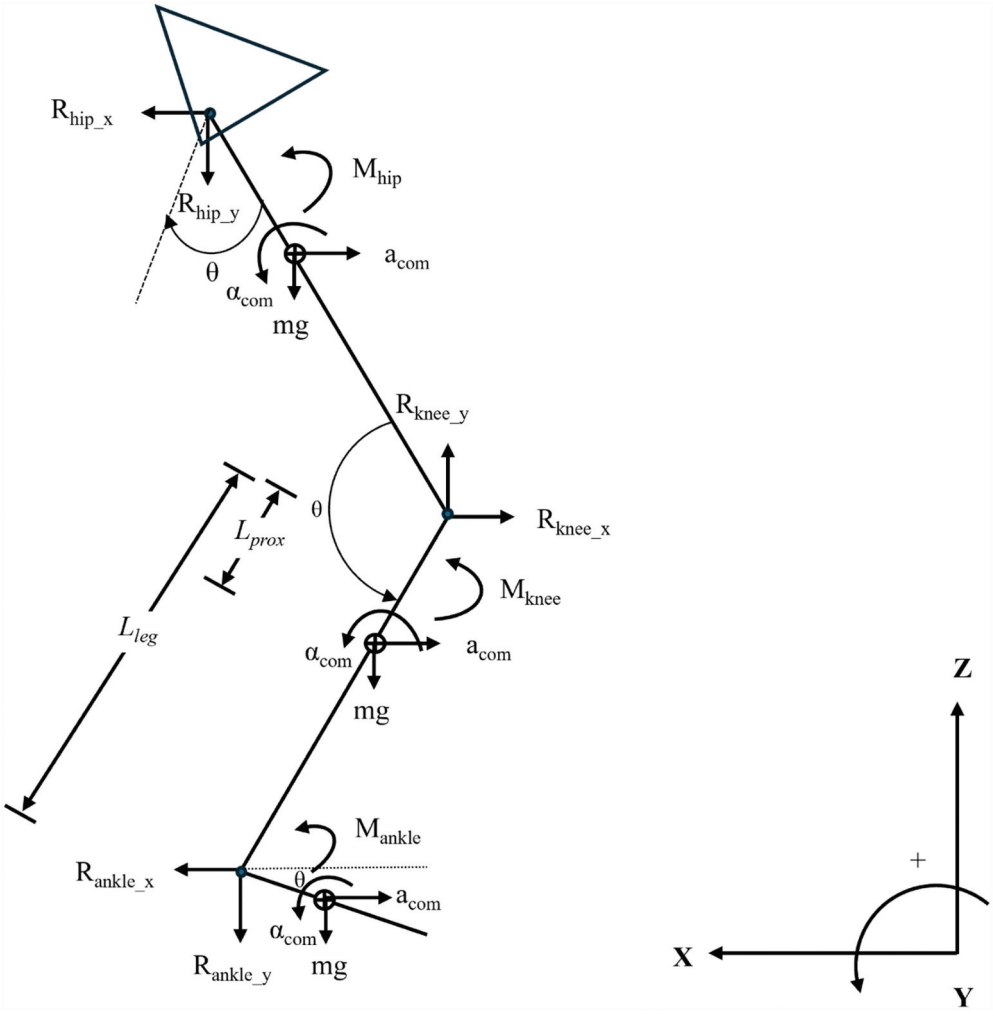

Figure 1. Free body diagram of the lower limb for the inverse dynamic calculations.

where M_{knee} and M_{ankle} are the moments at the knee and ankle, respectively. I_{com} is the moment of inertia about the segment's centre of mass, α_{com} is the angular acceleration about the segment's centre of mass, a_{com} is the linear acceleration of the segment's centre of mass, m and g represent mass of the segment and gravity, respectively and R_{ankle_x} and R_{ankle_y} are the x- and y-components of the ankle's reaction force. θ is the joint angle, L_{leg} is the length of the lower leg segment, and L_{prox} is the length of the proximal portion of the segment. Joint powers for each joint (P_{joint}) were the three degree of freedom rotational powers between adjacent segments during ground contact and were calculated as follows:

$$P_{joint} = M_{joint}\omega_{joint}$$

where ω_{joint} is the joint's angular velocity. Data were then post-processed in a custom MATLAB® program (MathWorks, Natick, MA, USA) where ground contact was defined

as the period of foot-ground contact with the treadmill when the vGRF exceeded 40 N. The timing of negative and positive power was defined by identifying the point of zero crossing in the power curve, the transition from negative to positive power production. Values were normalised to body mass (W/kg).

Statistical analysis

Means, standard deviations, and confidence intervals were calculated for peak positive and negative ankle, knee, hip, and total power for both limbs normalised to the participant's body mass. Then, to obtain the relative contribution of each joint's positive power to the total calculated, we divided the average power of each joint by the sum of the average positive power of the hip, knee, and ankle. This was repeated for negative powers. We screened the data for normality using a Shapiro-Wilk test. Results showed all knee power values for both limbs departed significantly from normality, $p \leq .05$. Therefore, the non-parametric test, Wilcoxon Signed Rank was used to determine the mean differences in the joint power variables between the pre- and post-test runs, $p \leq .05$. These statistical calculations were conducted using SPSS (ver. 29). Effect sizes (r) were also calculated for the pairwise comparisons using the formula, $r = \frac{|Z|}{\sqrt{N}}$ (Pallant, 2020). An r effect size of 0.1 was considered small, 0.3 a medium effect, and 0.5 a large effect (Cohen, 2013).

Results

Figures 2 and 3 present mean and standard deviation values for pre- and post-test total limb positive and negative power, respectively. There were no significant changes in total positive power for left ($Z = -0.073$, $p = .942$, $r = .01$) or right ($Z = -0.419$, $p = .675$, $r = .06$) limbs. Negative total power decreased for both limbs, and this was statistically significant for the left ($Z = -2.113$, $p = .035$, $r = .30$) but not the right ($Z = -1.776$, $p = .076$, $r = .25$) limbs.

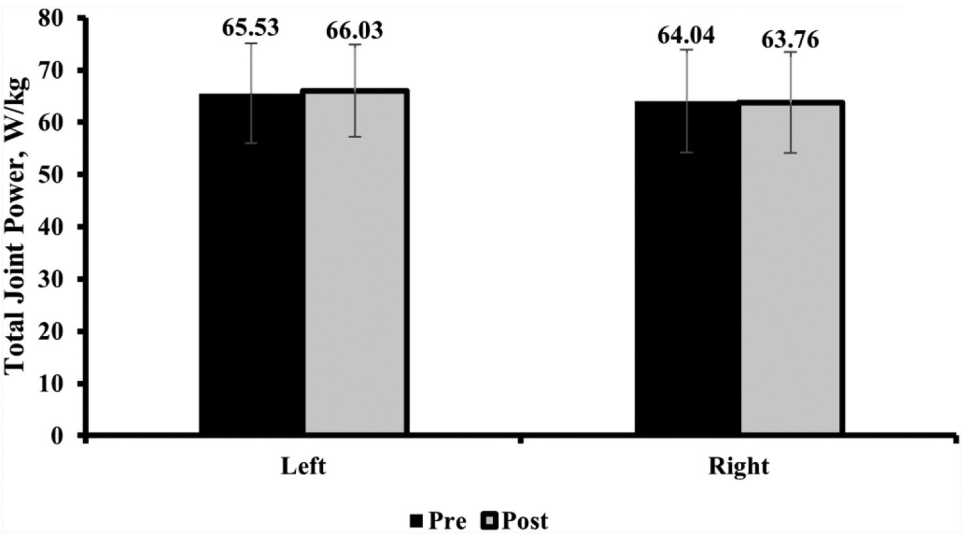

Figure 2. Pre- and post-test positive total joint power for left and right limbs.

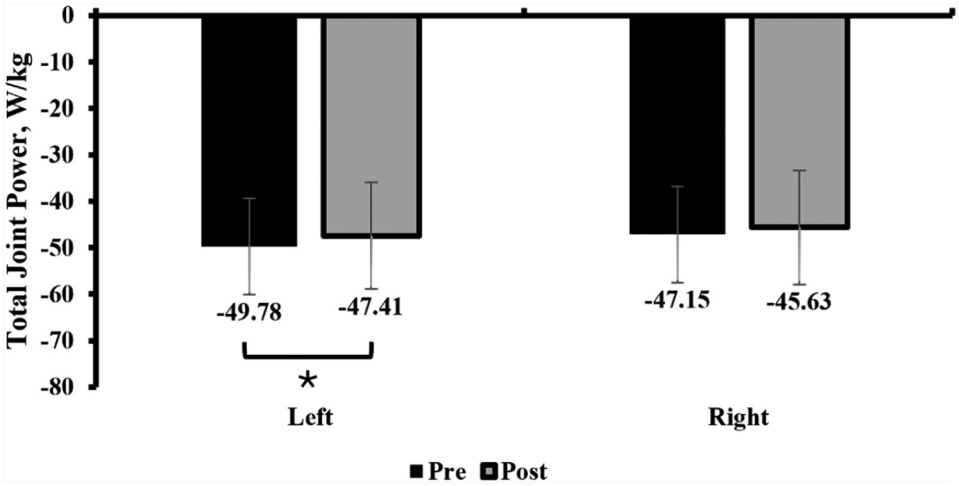

Figure 3. Pre- and post-test negative total joint power for left and right limbs. *denotes significant difference, $p < .05$.

Table 2 presents the means, standard deviations, 95% confidence intervals (CI), p levels, and effect sizes for positive and negative individual joint and power for left and right limbs pre- and post-training camp. Negative ankle joint power significantly decreased for the left ($Z = -1.967$) and right ($Z = -2.623$) limbs, $p < .05$. No other significant changes occurred in positive or negative hip or knee joint powers ($p > .05$).

Table 3 presents the means, standard deviations, 95% confidence intervals (CI), p levels, and effect sizes of the relative contributions of positive and negative joint power to total power for left and right limbs. Figures 3 and 4 depict the pre- and post-test relative contribution of the ankle, knee, and hip to the total positive and negative power for both limbs.

While there was a general shift in positive joint power from distal-to-proximal for both limbs (see Figures 3 and 4), these changes were not statistically significant. There were no significant changes in the percent contribution for the hip (left, $Z = -1.941$; right, $Z = -1.640$), knee (left, $Z = -1.256$; right, $Z = -0.530$), or ankle (left, $Z = -0.914$); right, $Z = -0.873$) joints after six weeks of training, $p > .05$.

For negative joint power contribution, there was also a shift from distal-to-proximal (see Figures 4 and 5), and this was significant on the left limb. Specifically, the left hip significantly changed (increased) ($Z = -2.398$,), $p < .05$. There were no significant changes in the right hip ($Z = -0.595$) and both knees (left, $Z = -0.628$; right, $Z = -1.126$) and ankles (left, $Z = -1.534$; right, $Z = -1.142$) after six weeks of training camp, $p > .05$.

Discussion and implications

We aimed to identify the joint power contribution changes during a speed running bout in American style football players before and after participating in a six-week NFL draft preparation camp. The camp included focused drills for developing linear speed (Table 1) as well as resistance training and position specific drills. Players were highly skilled with

Table 2. Means, SD, 95% CI, *p* levels, and effect sizes of pre- and post-training camp individual joint power for left and right limbs, $N = 52$.

	Side	Test	Mean ± SD (W/kg)	95% CI	*p* level	ES (*r*)
Positive						
Ankle	L	Pre	42.21 ± 7.33	(40.17, 44.25)	.854	.05
		Post	42.12 ± 7.34	(40.08, 44.17)		
	R	Pre	41.92 ± 7.90	(39.72, 44.11)	.255	.15
		Post	41.30 ± 7.62	(39.18, 43.42)		
Knee	L	Pre	8.31 ± 3.30	(7.39, 9.23)	.262	.15
		Post	8.12 ± 3.86	(7.05, 9.19)		
	R	Pre	7.52 ± 3.12	(6.65, 8.39)	.639	.04
		Post	7.31 ± 3.15	(6.43, 8.19)		
Hip	L	Pre	15.01 ± 3.62	(14.00, 16.02)	.183	.15
		Post	15.79 ± 4.32	(14.59, 16.99)		
	R	Pre	14.60 ± 3.42	(13.65, 15.55)	.150	.16
		Post	15.15 ± 3.99	(14.04, 16.26)		
Negative						
Ankle	L	Pre	−24.53 ± 7.53	(−26.63, −22.44)	**.049***	**.27**
		Post	−22.93 ± 6.68	(−24.79, −21.07)		
	R	Pre	−24.39 ± 7.82	(−26.57, −22.22)	**.009***	**.36**
		Post	−22.67 ± 7.39	(−24.73, 20.62)		
Knee	L	Pre	−14.40 ± 8.83	(−16.86, −11.95)	.101	.20
		Post	−13.62 ± 9.78	(−16.34, −10.90)		
	R	Pre	−12.33 ± 7.75	(−14.48, −10.17)	.940	.02
		Post	−13.09 ± 10.36	(−15.97, −10.20)		
Hip	L	Pre	−10.85 ± 3.84	(−11.92, −9.78)	.621	.01
		Post	−10.87 ± 3.38	(−11.81, −9.92)		
	R	Pre	−10.44 ± 3.32	(−11.36, −9.51)	.461	.16
		Post	−9.87 ± 4.29	(−11.07, −8.67)		

*denotes significant difference between pre- and post-training camp, $p \leq .05$.

35% invited to attend the prestigious combine and the remaining 65% performing for scouts and coaches at what are called 'pro days'. We hoped to gain insight into joint-specific mechanistic adaptations that would provide valuable information for what change is possible in high calibre team sport athletes attempting to enhance linear speed. Previous studies have assessed joint power redistribution during fatiguing or exhaustive runs (Melaro et al., 2021; Willer et al., 2021), changing stride mechanics and/or speed (Belli et al., 2002; Dewald et al., 2023; Schache et al., 2011), between recreational and competitive runners (Sanno et al., 2018), and in older runners (Kim & Park, 2022). However, to our knowledge no studies have documented differences before and after a training block in highly skilled athletes. We hypothesised that participation in training camp would redistribute both positive and negative joint power from the ankle towards the more proximal hip joint and knee joints even if treadmill speed were kept uniform between testing sessions. The hypothesis was partially accepted.

Since faster running speeds have been shown to change joint power contribution (Jin & Hahn, 2018, Schache et al., 2001; Schache et al., 2015), treadmill speed was controlled between testing sessions to aid in distinguishing camp participation effects on the dependent variables. The joint power magnitudes in the current study are comparable to those from similar speed runs in previous studies (Belli et al., 2002; Kariyama & Zushi, 2016) and show the ankle to be the dominant power producer bilaterally for both absorption (~50%) and release (~65%) (Belli et al., 2002; Schache et al., 2015; Stearne et al., 2014). Total positive power did not change for both limbs after training camp.

Table 3. Means, SD, 95% CI, *p* levels, and effect sizes of pre- and post-training camp power contribution for left and right limbs, $N = 52$.

	Side	Test	Mean ± SD (%)	95% CI	*p* level	ES (*r*)
Positive						
Ankle	L	Pre	64.40 ± 6.35	(62.63, 66.17)	.261	.10
		Post	63.90 ± 8.01	(61.67, 66.13)		
	R	Pre	65.32 ± 6.47	(63.52, 67.12)	.383	.09
		Post	64.73 ± 7.13	(62.74, 66.71)		
Knee	L	Pre	12.62 ± 4.41	(11.40, 13.85)	.209	.20
		Post	12.12 ± 4.79	(10.79, 13.46)		
	R	Pre	11.62 ± 4.04	(10.50, 12.75)	.596	.09
		Post	11.38 ± 4.17	(10.22, 12.54)		
Hip	L	Pre	22.97 ± 4.54	(21.71, 24.24)	.520	.24
		Post	23.98 ± 5.80	(22.36, 25.59)		
	R	Pre	21.62 ± 5.16	(21.62, 24.50)	.101	.20
		Post	23.89 ± 5.65	(22.32, 25.46)		
Negative						
Ankle	L	Pre	49.86 ± 12.72	(46.32, 53.40)	.125	.19
		Post	49.14 ± 11.81	(45.85, 52.43)		
	R	Pre	51.79 ± 13.53	(48.19, 55.39)	.253	.15
		Post	50.90 ± 12.99	(47.28, 54.52)		
Knee	L	Pre	28.62 ± 12.58	(24.95, 31.80)	.530	.08
		Post	27.84 ± 11.04	(24.76, 30.91)		
	R	Pre	25.86 ± 11.84	(22.56, 29.16)	.260	.13
		Post	27.44 ± 13.11	(23.79, 31.09)		
Hip	L	Pre	21.77 ± 6.16	(20.05, 23.48)	**.016***	**.31**
		Post	23.02 ± 5.75	(21.42, 24.62)		
	R	Pre	22.35 ± 6.21	(20.62, 24.08)	.552	.07
		Post	21.66 ± 7.41	(19.60, 23.72)		

*denotes significant difference between pre- and post-training camp, $p \leq .05$.

Interestingly, while the magnitudes of positive ankle power showed virtually no change at the post-run, there was a minor reduction (~0.5 %) in positive ankle contribution to total limb power that appears to be the result of small mean increases in positive hip power magnitude (0.55–0.75 W/kg) that shifted the contribution. This is an intriguing finding since hip power has been shown to increase with cumulative running speeds above ~7.0 m/s (Dorn et al., 2012, Schache et al., 2001; Schache et al., 2015). The mean run test speed in the current study was 6.4 m/s and was the same for both testing sessions. Muscles around the hip appear to have generated greater work during the propulsion phase (Figure 6) just as occurs with running at faster speeds. This may be a marker of improved limb efficiency. However, the effect sizes were small (left, $r = .24$; right, $r = .20$). Further, not all participants increased positive hip joint power. Ten players (19%) decreased by 0.5 to 6.0 W/kg. A runner may become more reliant upon hip muscles if unable to generate power through the calf muscles (van Oeveren et al., 2024). This has been shown to occur when runners shift to a higher step frequency at a lower speed (van Oeveren et al., 2024) which also lowers ground contact time. Subjects in the current study decreased their ground contact times on the left side from 160 ms to 158 ms and maintained 158 ms on the right side at both testing runs leading to symmetrical ground contact times. Limb symmetry has been the subject of biomechanical analyses of sprint (Exell et al., 2017; Exell et al., 2012; Kaçoğlu, 2019) and distance (Gao et al., 2020, Hashizume et al., 2019; Vial et al., 2023) runners. Exell et al. (2017) examined relationships between various

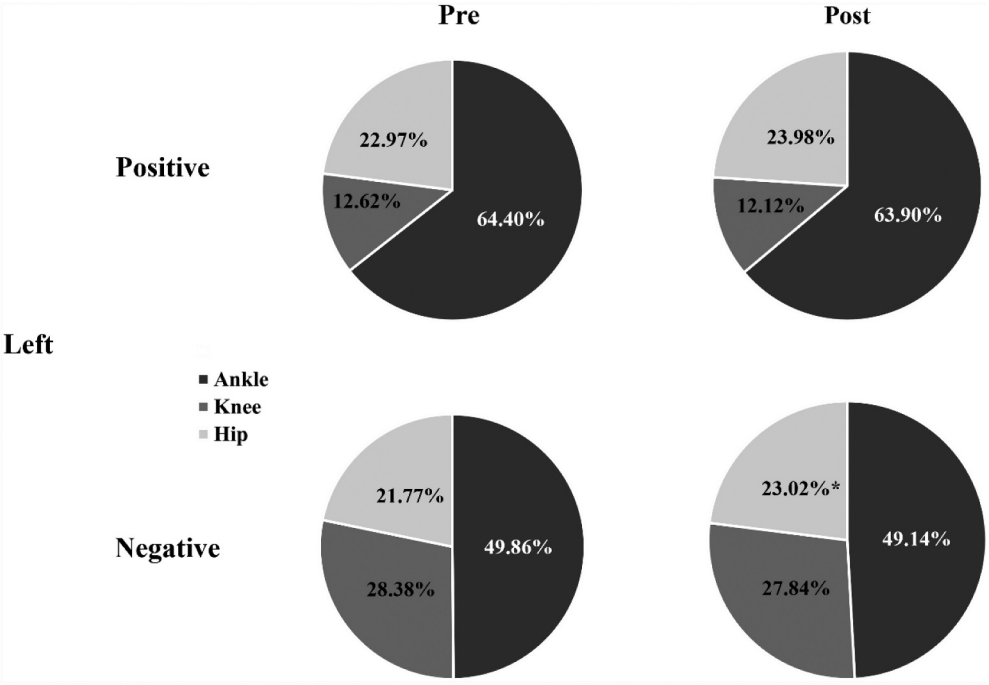

Figure 4. Relative contribution of the left ankle, knee, and hip to the total average positive and negative joint power during the pre-test (top) and post-test (bottom). The radius of each pie chart is scaled based on the total positive power of each condition. *denotes significant difference, $p < .05$.

kinetic, kinematic, strength (squat jump GRFs), and performance asymmetries in eight male competitive sprinters. Results showed the asymmetries were highly individual and there were no significant relationships except for ankle work during sprint running and peak vertical force ($r = .895$) and power ($r = .761$) during the squat jumps. The researchers suggested that the ankle joint may be vital in regulating asymmetry in sprinting. Symmetry in joint power contributions were not examined in the current study. However, Table 3 shows minimal differences (~1–1.5%) in mean joint contribution values show between the right and left ankles. During sprinting, the ankle joint plays an important role in producing the necessary large vertical ground reaction force (vGRF) (Dorn et al., 2012; Schache et al., 2014; Stefanyshyn & Nigg, 1998), reducing ground contact time (Charalambous et al., 2002), and enhancing mechanical efficiency (Kuitunen et al., 2002). Specifically, the ankle plantar flexor muscles (gastrocnemius and soleus) have a dominant role in those tasks and rely greatly on the utilisation of tendon elastic strain energy for generating power during the stance phase. In the current study, mean stance-averaged vGRFs were unchanged for the left (pre, 18.39 ± 1.38 vs. post, 18.47 ± 1.33 N/kg) and right (pre, 18.37 ± 1.78 vs. 18.28 ± 1.61 N/kg) limbs which seems congruent with the stability of the magnitudes of positive total limb power.

Total limb negative power decreased after training and was statistically significant for the left limb with a medium effect size ($r = .30$). The reduction may indicate an improved ability to absorb energy in the initial part of stance. Significant decreases in negative ankle power magnitude influenced this finding. Most of the power absorbed relative to the total

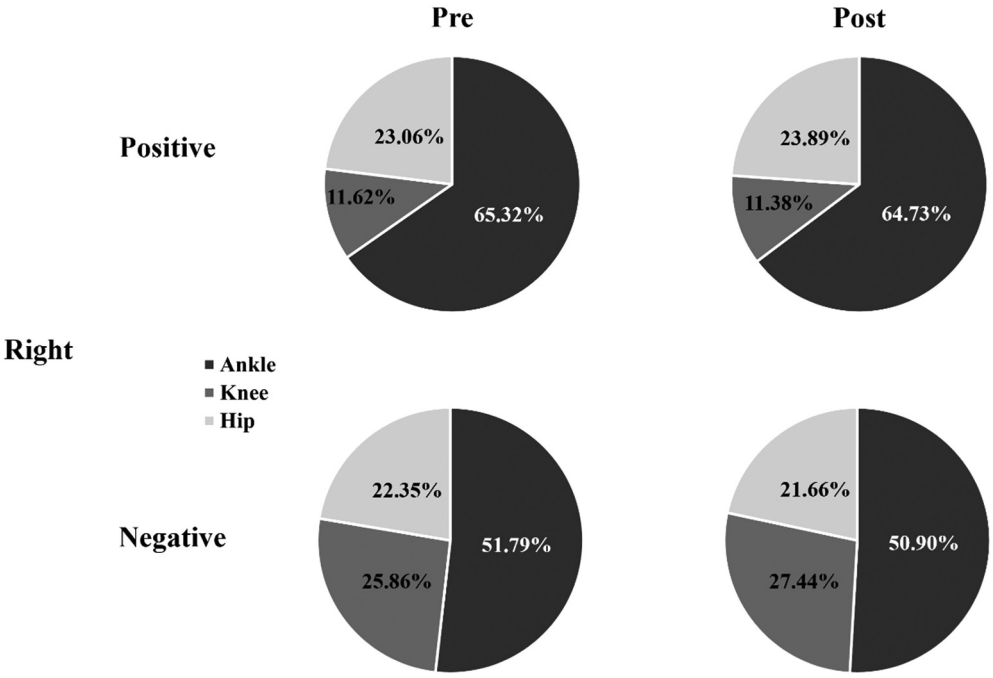

Figure 5. Relative contribution of the right ankle, knee, and hip to the total average positive and negative joint power during the pre-test (top) and post-test (bottom). The radius of each pie chart is scaled based on the total positive power of each condition.

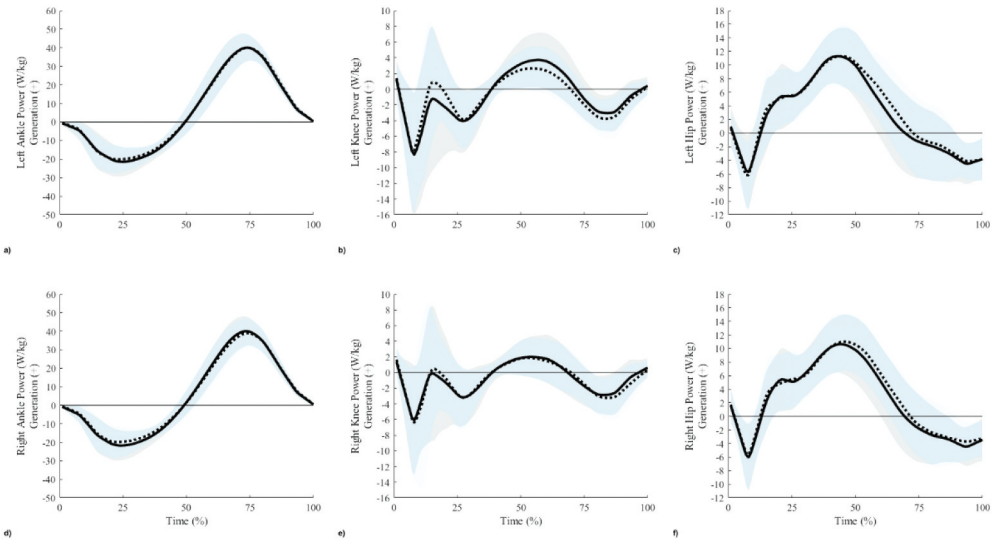

Figure 6. Time series normalised to stance phase (100%) for left (top row) ankle (a), knee (b), and hip (c) power and right (bottom row) ankle (d), knee (e), and hip (f) power for the pre-test and post-test.

limb power (~47–51%) at both testing sessions was performed around the ankle joint. Greater muscle damage occurs with cumulative negative work, or eccentric loads (Hashizume et al., 2019). Thus, this finding was seen as not only potentially enhancing for performance but also increasing injury protection. Further, ground contact times have been found to be relatively consistent up to 20 km/hr and then may vary with lesser skilled runners demonstrating an inability to generate high braking and propulsive forces in a short time by the muscles. Participants increased hip and knee negative power during braking period and increased hip positive power during propulsion. Enhancements in swing leg mechanics, specifically 'leg retraction' may have influenced the findings for negative joint power. Leg retraction is the rearward leg rotation before initial ground contact and is controlled primarily by hip rotation (Orendurff et al., 2018) with the extensors as prime forward movers of the body (Belli et al., 2002). Leg retraction enhances running mechanics by reducing horizontal velocity difference of the foot relative to the surface (Blazevich, 2017; Blum et al., 2010). This reduces impact impulse, prevents slipping (Karssen et al., 2015), improves stability (Blum et al., 2010; Karssen et al., 2011), and possibly decreases total limb negative power, specifically at the ankle. Decreases in negative power have been shown to result from improved muscle-tendon elasticity and stiffness (Hirayama et al., 2017; Kubo et al., 2010, 2012), stride mechanics (Swinnen et al., 2021), and neuromuscular control (Akbar et al., 2022). Thus, this change may be a training adaptation reflective of enhanced linear speed mechanics. However, all players did not exhibit this shift. Seventeen (~33%) participants had 0.5 to 9 W/kg increases in negative hip joint power and five (~10%) had ≥ 9 W/kg increases in negative ankle joint power. Intervention programs of ≥12 weeks have been shown to be most beneficial to muscle-tendon adaptation (Bohm et al., 2015; Hirayama et al., 2017) while stride mechanics and neuromuscular control programs may induce beneficial changes in performance in 2–8 weeks (Akbar et al., 2022; Barber-Westin et al., 2015; Maćkała & Fostiak, 2015). Thus, it is plausible that the sprint technique drills, and the other related camp exercises may have reduced the need of the ankle joint's muscle-tendon complex for energy absorption for most of the participants. To illustrate, the first five linear speed drills in Table 1 focus on rapid ankle dorsiflexion at foot-off and piston-like motion at the hips and knees. Improvements in force application mechanics may have aided in redistributing joint power especially for players of positions where sprinting is required such as offensive and defensive linemen. However, position differences were not evaluated in the current study. This group began training camp with most participants just finishing their competitive playing season where mechanics of game specific demands such as changes of direction and large decelerations were predominant for most player positions. The goal of training camp was to prepare participants for their professional try-out in which they showcase linear speed in the 36.6 m run.

The unique participant pool, elite-level American style football players training for the NFL draft at a single facility presented an opportunity to monitor a highly specialised sample but also made it difficult to utilise a control group. Without a control group, we cannot be confident that the observations were solely from the training camp protocol. Footwear differences between testing sessions, muscle soreness, fatigue, energy intake, limb dominance, or previous injury may have influenced the findings. However, it should be noted that training camp protocols for resistance training, field-drills, sleep, nutritional supplements, recovery

treatment, corrective exercises, and injury management are strict and consistent strengthening findings within pre-post test limitations. Furthermore, 43/52 (83%) of the participants were categorised as Skill (e.g., wide receivers, defensive backs) or Big-skill (e.g., tight ends, linebackers) who run the ball more in American style football. The remaining were Big players (linemen) who primarily block and tackle. Thus, the small to moderate *r* effect sizes observed may have been influenced by player position.

Conclusion

This study is the first to longitudinally examine changes in joint-level power magnitude and contribution during speed running before and after a training block in a group of highly skilled athletes. The goal was to gain insight into joint-specific mechanistic adaptations that would provide valuable information for what is possible in high calibre team sport athletes attempting to enhance linear speed. The primary generator of both positive and negative joint power was the ankle, and this was consistent with previous findings (Swinnen et al., 2021). Positive power contribution at the ankle decreased by ~ 0.5% and increased at the hip by ~ 1.0%, but these were not significant. The interesting change was that of decreased negative ankle power bilaterally indicating less reliance on the ankle's muscle-tendon complex for energy absorption during the first portion of the stance phase. This change occurred despite running at the same speed (mean 6.4 m/s) before and after training camp. The distal-to-proximal shift in negative joint power as a by-product of training requires further investigation. Given that the high skill of the participants may affect the sensitivity of change of the variables in this study, the moderate and low effect sizes produced may still warrant attention. We also observed individual differences in the training response that might have affected the results. Further, future researchers may consider examining adaptations between player position groups. We suggest that the American style football players in the current study may have adopted the joint power mechanics of those exhibited when running at faster speeds and this could be considered a training attribute.

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